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-- DELHIVERY LOGISTICS & SUPPLY CHAIN DASHBOARD

-- ADVANCED DAX MEASURES

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-- 1. WEATHER DELAY RATE

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Weather Delay Rate =

DIVIDE(

 CALCULATE(

 [Total Shipments],

 Delivery_Logistics[delayed] = "Yes"

),

 [Total Shipments],

 0

)

-- =====

-- 2. SHIPMENT STRESS INDEX

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Shipment Stress Index =

VAR DelayPressure =

[Delay Rate]

VAR CostPerKM =

DIVIDE(
SUM(Delivery_Logistics[delivery_cost]),
SUM(Delivery_Logistics[distance_km]),
0
)

VAR OverallCostPerKM =

CALCULATE(
DIVIDE(
SUM(Delivery_Logistics[delivery_cost]),
SUM(Delivery_Logistics[distance_km]),
0
,
REMOVEFILTERS(Delivery_Logistics)

)

VAR CostPressure =

DIVIDE(

CostPerKM,

OverallCostPerKM,

0

)

VAR AvgDistance =

AVERAGE(Delivery_Logistics[distance_km])

VAR OverallAvgDistance =

CALCULATE(

AVERAGE(Delivery_Logistics[distance_km]),

REMOVEFILTERS(Delivery_Logistics)

)

VAR DistancePressure =

DIVIDE(

AvgDistance,

OverallAvgDistance,

0

)

VAR AvgWeight =

AVERAGE(Delivery_Logistics[package_weight])

VAR OverallAvgWeight =

CALCULATE(

AVERAGE(Delivery_Logistics[package_weight]),

REMOVEFILTERS(Delivery_Logistics)

)

VAR WeightPressure =

DIVIDE(

AvgWeight,

OverallAvgWeight,

0

)

RETURN

(

MIN(DelayPressure, 1) * 40

)
 +
 (
 MIN(CostPressure, 2) / 2 * 25
)
 +
 (
 MIN(DistancePressure, 2) / 2 * 20
)
 +
 (
 MIN(WeightPressure, 2) / 2 * 15
)

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-- 3. OVERALL PERFORMANCE SCORE

-- =====

Overall Performance Score =

VAR CompanyCost =

[Avg Delivery Cost]

VAR CompanyRating =

[Avg Rating]

VAR CompanyEfficiency =

[Cost Efficiency]

VAR MinCost =

**MINX(
ALL(Delivery_Logistics[delivery_partner]),
[Avg Delivery Cost]
)**

VAR MaxCost =

**MAXX(
ALL(Delivery_Logistics[delivery_partner]),
[Avg Delivery Cost]
)**

VAR MinRating =

**MINX(
ALL(Delivery_Logistics[delivery_partner]),**

[Avg Rating]
)

VAR MaxRating =

MAXX(
ALL(Delivery_Logistics[delivery_partner]),
[Avg Rating]
)

VAR MinEfficiency =

MINX(
ALL(Delivery_Logistics[delivery_partner]),
[Cost Efficiency]
)

VAR MaxEfficiency =

MAXX(
ALL(Delivery_Logistics[delivery_partner]),
[Cost Efficiency]
)

VAR CostScore =

```

IF(
    MaxCost = MinCost,
    0.5,
    DIVIDE(
        MaxCost - CompanyCost,
        MaxCost - MinCost,
        0
    )
)

```

VAR RatingScore =

```

IF(
    MaxRating = MinRating,
    0.5,
    DIVIDE(
        CompanyRating - MinRating,
        MaxRating - MinRating,
        0
    )
)

```

VAR EfficiencyScore =

```

IF(
    MaxEfficiency = MinEfficiency,
    0.5,
    DIVIDE(
        CompanyEfficiency - MinEfficiency,
        MaxEfficiency - MinEfficiency,
        0
    )
)

```

```

RETURN
(
    CostScore * 35
)
+
(
    RatingScore * 35
)
+
(
    EfficiencyScore * 30
)

```

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-- 4. OVERALL PERFORMANCE RANK

-- =====

Overall Performance Rank =

RANKX(

ALL(Delivery_Logistics[delivery_partner]),

[Overall Performance Score],

,

DESC,

DENSE

)